Proceedings of the IASTED International Conference
ARTIFICIAL AND COMPUTATIONAL INTELLIGENCE
September 25-27, 2002, Tokyo, Japan

EFFECTIVENESS OF ACTION PREDICTION METHOD FOR A USER USING INDUCTIVE LEARNING WITH N-GRAM

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ABSTRACT

This paper describes a method for action prediction of a user. When we build a care system with learning function like a learning room, a statistical approach or an analytical approach can be considered. Statistical approaches are not liable to produce reliable result unless a huge prepared database is available. The analytical approaches are necessary to give the prepared rules adapted to the user and the adaptability of this method is low. Aiming at solution of such problems, we have proposed a method to predict the action of a user using Inductive Learning with N-gram. The system based on this method is able to acquire needed rules from comparative few data history automatically using Inductive Learning. The rules express a user's taste and custom. Therefore the system is able to adapt dynamically to the users by it's own learning ability. The rate of the average correct prediction was 60.1[%] on the experiment. The user must proofread the erroneous conversion in the prediction results. However, the erroneous conversion decreases since the system based on this method is able to adapt dynamically to various users. This paper shows the evaluation results of the action prediction in our proposed method.

KEY WORDS:

Inductive Learning, N-gram, Action prediction, Learning Room, Caring system, Adaptability

1. INTRODUCTION

Recently, baby birthrate is decreasing and it seems that the society is aging. Caring for the aged or bodily inconvenient persons is under pertinent requirement. However, not everyone can necessarily afford to employ a person looking after a patient. For the aged or bodily inconvenient persons a caring system instead of manually_care needs to be developed. In case such a system is developed, the most important problem is how to predict a user's intention and action. It is clear that for each user the intention is different from action and it is difficult to give prepared knowledge for the system. Accordingly, predicting the intention from a user's actual action's history by Inductive Learning is desirable.

Inductive Learning is an acquired inherent rule through extracting a common part and a difference part recursively from examples [1]. The natural language processing technology using Inductive Learning is a research aiming at elucidation and its engineering application of human being's learning capacity [2]. In such research, in order to solve human being's learning capability, knowledge cannot be given to a system beforehand at all, but only the knowledge that the system acquired from the examples is used and the processing is advanced, and the effectiveness is proved [3].

On the other hand, recently, technology of ID and position recognition is greatly improved in the authentic situation of recognition. Therefore, the recognized information can be processed using Inductive Learning. Thus, using the Inductive Learning from the data of human's behavior and his states we can obtain rules of his custom and tastes. Then the system becomes to be able to choose own behavior suitable for the specific user. Such personalized data could help to confirm the specific user's intention implicitly and be useful as triggering element for interaction with the user, since the action information and the system's choices should be indicated. Accordingly it is useful to decide the cause that starts interaction [4].

Taking advantage of such natural language processing technology, we discuss the realization possibility using Inductive Learning. We suppose that the care system is a learning room. This room's states can be acquired now by progress of technology. The care system has knowledge acquisition capability that can learn various kind of information from an environment and grows up.

In order to give such a function to the system, the state of the room united with a user's action or the rules that can guess intentions of a user is needed.

Various learning approaches are planned when we build a learning room system. To use the analytical approach needs to give the prepared rules. The problem is that the custom and taste will change with different users. Moreover the rules that adapt to each user are needed. Consequently dynamic adaptation of a system cannot be realized. Likewise, we can also consider it from statistical approach, for example, Whole Learning, Expectation Maximization Algorithm and Bayesian Learning Algorithm and so on [5] that we may plan to use. The following problems may come out. The Whole Learning Algorithm is a method that the prepared data sets are given to the system, and then realize the learning function. Therefore it needs the huge database of user's action in order to acquire a good prediction result. As to the Expectation Maximization Algorithm, since the prediction accuracy depends on the size of prepared database, a better prediction result is not obtained while using the learning-room system. The Bayesian Learning Algorithm calculates for a prediction distribution. Similarly, a huge database is also required for the better prediction result. Moreover, with the action changing with users, it is difficult to give a lot of examples beforehand for each user. It is a problem to use these statistical approaches so that it involves a lot of time and heavy labor preparing so many examples for each user. For the others, it is a fault that the exception examples cannot be coped with these methods.

In order to solve such problems, we introduce the Inductive Learning. Following merits appears: The first, the characteristic of Inductive Learning is that it can be learned with sufficient accuracy even from a small number of data. It is possible to acquire the rules including the exception example. The next, because it is learned from the state of the learning room while the system can be acquired automatically, it is unnecessary to make a huge database for each user beforehand. Moreover, because the data is different with each user, the extracted rules will be different so that the dynamic adaptation is possible to each user. In short, this method is robust. The system based on it can adapt to a user dynamically by acquiring rules from examples using the Inductive Learning [1].

The Inductive Learning can estimate the likelihood of each rule. In order to evaluate the whole relations of the rules, we introduce the N-gram used to our system. We define this method is the action prediction of a user using Inductive Learning with N-gram. If we consider from an action sequence of a user, it is required to analyze relation of action in alignment with the passage of time for the system. On the other hand, it is the script, which expresses the standardized action that a person performs ordinary things in the form of the chain of a phenomenon in alignment with the passage of time [6]. It is considered from the analysis of a script using N-gram that a user's action can be predicted [7]. We propose this method is the action prediction of a user using Inductive Learning with N-gram. This paper outlines this method and the experimental results for the evaluation of its adaptability. The system based on this method generates a data history and a rule dictionary adapted to a target user by Inductive Learning. We consider that this method is effective for a device such as a care system like the learning room.

In the following chapters, the component of a system and the outline of this method are explained. Hereafter, a result is described as the experiment method and consideration of a result is performed. Finally future works are described.

2. DESIGN OF THE SYSTEM (LEARNING ROOM) AND THE FUNDAMENTAL CONCEPT

The system based on this method is assumed to be a learning room for realizing a care system. For easy analysis, the component of the room consists of eight artificial elements, such as a door (D), TV, TV program (P), curtain(C), light (L), an air-conditioner (AC), a telephone (TEL), and a bell (B). Not only the state of an artificial element but also the state of natural elements, such as time, temperature and brightness are included. The changed coefficient of the state shows what a user does before. It is assumed that the system can acquire the state of the room automatically with actions of the user.

2.1 DESCRIPTION OF CAUSAL RELATIONSHIP

Natural element (N) consists of time, brightness, day of week, and temperature in this system. Natural element is symbolized as follows:

$$N_k^n = | W_k^n | H_k^n | M_k^n | B_k^n | T_k^n |$$ *1

Similarly, artificial element (ART) is symbolized as follows:

$$ART_k^n = | D_k^n | TV_k^n | P_k^n | C_k^n | L_k^n | AC_k^n | TEL_k^n | BELL_k^n |$$

When a user carries out an action, the state of a certain artificial element changes according to this user's operation. The result which changed the state of the

1 : n expresses the order of the data registered into the state history. K expresses the date. | is a symbol to distribute about the state of each element。

Table 1: Setting Value of Each Element

ELEMENT	SETTING VALUE						
Day of Week	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Time	hh & mm						
Brightness	50lux	100lux	150lux	...	1400lux	1450lux	1500lux
Temperature	16°C	17°C	18°C	...	33°C	34°C	35°C
Door	Doff				Don		
TV	Toff	1ch	2ch	...	58ch	59ch	60ch
TV-program	None	Sport	News	Music	Film	Cartoon	Economics
Curtain	Coff				Con		
Light	Loff				Lon		
Air conditioner	Aoff	16°C	17°C	...	33°C	34°C	35°C
Telephone	Teoff		Ring		Teon		
Bell	Boff				Bon		

artificial element can be treated as a user's reaction(REA_k^n). The state of the room unites the state of natural elements and artificial elements. It is considered that the state of the room that is just before the user taking a certain action, is the cause used as the action of the user. Thereupon, the cause used as action of a user is defined as follows:

$C_k^n = N_k^n \,||\, ART_k^n$ *2

The result(REA_k^n) of changed state of a user's action and a corresponding artificial element reacts to an intention of a user. Using this result, we can find the causal relations such as the $C_k^n \rightarrow REA_k^n$ exists. We define $C_k^n \rightarrow REA_k^n$ as a rule with causal relation.

2.2 THE SETTING VALUE FOR EACH ELEMENT OF THE SYSTEM

In order to express the state of the learning room in this system, each element has some values given beforehand. Table 1 shows the setting values.

2.3 ACQUISITION OF A RULE WITH CAUSAL RELATIONS

Of the definition of an initial rule, it is the most concrete rule that has the correspondence relation ($C_k^n \rightarrow REA_k^n$) extracted from the data history. The variables are not included in the rules. For example, acquisition of the rule that TV will be turned off if a telephone is taken.

The example of data:

S_1^1 =|Monday|7:31|750|24||Doff|3ch|News|Con|Loff|Aoff|**Ring**|Boff|

S_1^2 =|Monday|7:31|750|24||Doff|**3ch**|**News**|Con|Loff|Aoff|**Teon**|Boff|

S_1^3 =|Monday|7:31|750|24||Doff|**Toff**|**None**|Con|Loff|Aoff|Teon|Boff|

Here, the common parts and the difference parts between S_1^1 and S_1^2 can be extracted, then an initial rule ($S_1^1 \rightarrow$Teon) can be acquired. Similarly ,from S_1^2 and S_1^3 the rule($S_1^2 \rightarrow$|**Toff**|**None**|) can be acquired, too. The difference parts and the common parts of the initial rules can be extracted by repeating for abstracting the rules. For example, when the data is given such as:

S_2^1 =|Tuesday|10:03|800|28||Doff|1ch|News|Con|Loff|Aoff|**Ring**|Boff|

S_2^2 =|Tuesday|10:03|800|28||Doff|**1ch**|**News**|Con|Loff|Aoff|**Teon**|Boff|

S_2^3 =|Tuesday|10:03|800|28||Doff|**Toff**|**None**|Con|Loff|Aoff|Teon|Boff|

The initial rule which($S_2^2 \rightarrow$|**Toff**|**None**|) can be extracted from S_2^2 and S_2^3 .Then the rule that had been abstracted from the rule ($S_1^2 \rightarrow$|**Toff**|**None**|) and the rule($S_2^2 \rightarrow$|**Toff**|**None**|) is:

|@0|@1|@2|@3||Doff|@4|**News**|Con|Loff|Aoff|**Teon**|Boff|
$\rightarrow$|**Toff**|**None**|

In this formula, @0, @1, @2, @3, and @4 are the variables in the abstracted rule.

3. OUTLINE OF SYSYTEM

Figure 1 shows an outline of the system based on this method. The procedure consists of the prediction process, the correction process, the feedback process and the learning process in this order.

*2: "||" is a symbol to distribute about a classification of each symbolized element.

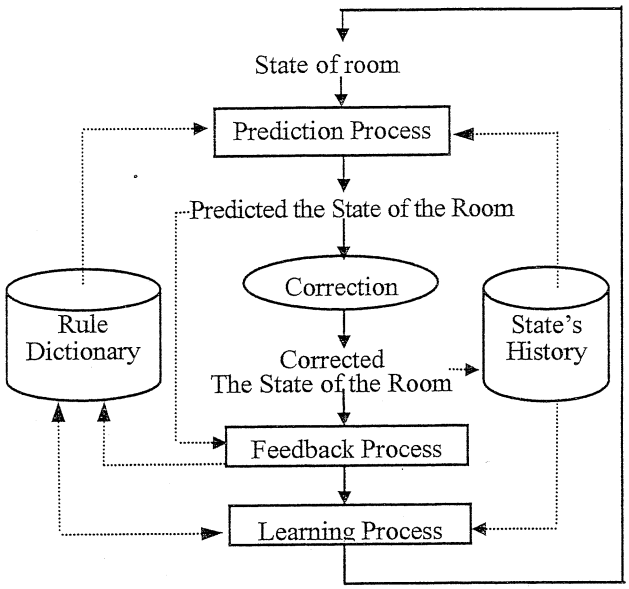


Figure 1: Procedure

The state of the room where it was inputted performs prediction processing with reference to the state history and the rule dictionary by pattern matching. The predicted state of the room progresses to feedback processing when it is right. Otherwise, when it has errors, the correction process is performed. The user judges whether it is correct or not and corrects them, the corrected state of the room registers into the state history, and it progresses to feedback processing. The rules in the rule dictionary are acquired in the learning process, and it is classified into four ranks which are M,H,C and L. The rules are applied in this order. The rank of the rule dictionary is shown in Table 2.

Table 2.: The Rank of the Rule Dictionary

NAME OF RANK	THE RATE OF THE CORRECT PREDICTION
Most (M)	CR≥90%
High (H)	70%≤CR< 90%
Common(C)	30%≤CR< 70%
Least (L)	CR< 30%

Increases in efficiency of a dictionary and an activation are attained. CR and ER are used also for the credibility evaluation function. The credibility of the rules is evaluated by the credibility evaluation function if there are some candidates of the rule in the same rank. The credibility evaluation function CEF is defined as:

$$CEF = \alpha P(A_{n+1} | A_n) + \beta CR - \gamma ER \quad \bullet\bullet\bullet (1)$$

,where α, β and γ are coefficients. CR is the rate of the correct prediction. ER is the rate of the erroneous prediction. $P(A_{n+1} | A_n)$ is the probability of the neighboring action. CR and ER are based on the rule dictionary and they are updated in the feedback process. $P(A_{n+1} | A_n)$ is based on the state's history. The credibility for the rule is higher when $P(A_{n+1} | A_n)$ is higher, and CR is higher and ER is lower. The CR and ER defined as (2) and (3):

$$CR = \frac{CF}{CF + EF} \quad \bullet\bullet\bullet (2)$$

$$ER = \frac{EF}{CF + EF} \quad \bullet\bullet\bullet (3)$$

,where CF is the appearance degree for the correct prediction, EF is the appearance degree for the erroneous prediction. In the feedback process, the predicted state of the room is compared with the corrected state of the room. According this result, the credibility of the rules, CF, EF and CR. And rules rank are updated. The certainty degree for the rule in the rule dictionary is updated, too. When a result of the prediction is correct, CR of the rule increases because its certainty degree increases. When a result of the prediction is erroneous, its ER increases because its certainty degree decreases. It is judged by comparing the result of prediction and its corrected result. In the learning process, the extraction of the initial rule is performed from the data history. The extracted initial rule is registered into the rule dictionary. Moreover, A rule is abstracted by the repetition of extracting the common and the difference parts of the initial rule. The abstracted rule is registered into the rule dictionary. In the case, that recursive processing of a rule is performed, it is necessary to give the suitable degree of abstraction. In this system, the definition of the rule abstraction degree is shown in the formula (4) according to the feature that the natural element and the artificial element have each parameter:

$$DA = (\frac{NVNE}{NNE} + \frac{NVAE}{NAE}) \times 100\% \quad \bullet\bullet\bullet (4)$$

,where DA is the abstraction degree of the rule. NVNE is the number of the variables for the natural element. NNE is the number of the natural element. NVAE is the number of the variables for the artificial element. NAE is the number of the artificial element in one abstracted rule. At the same time, from the data history, the script of action that the user performed for everyday is generated.

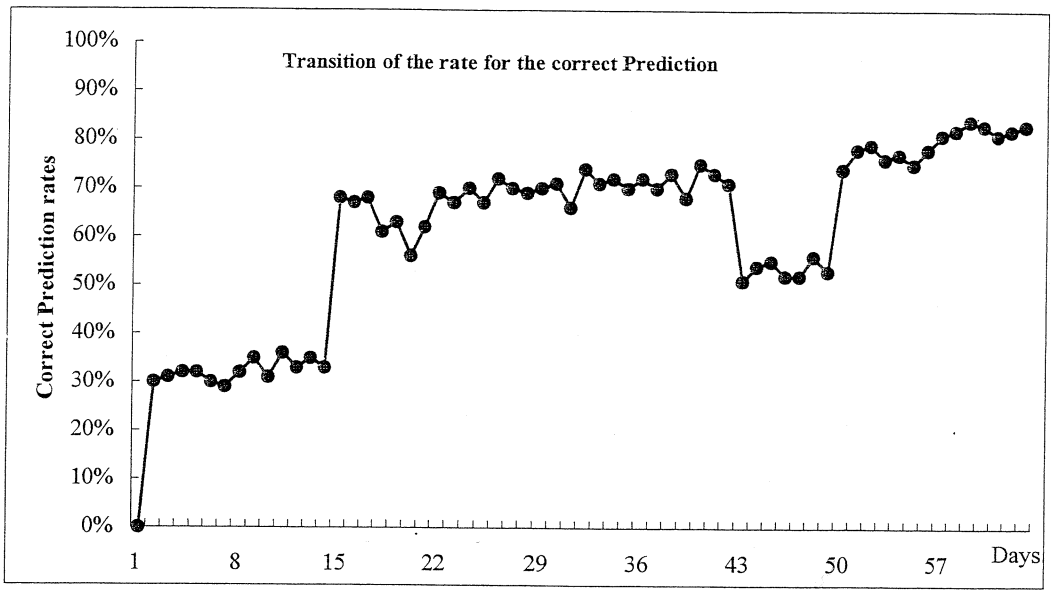


Figure 2 the rate of the Correct Prediction

The generated script is registered into the data history too. From the registered script, neighboring action probability $P(A_{n+1} \mid A_n)$ [*3] is analyzed by Bi-gram and the analyzed result is registered into the rule dictionary. Thus, the system based on this method is improved by the repetition of these processes.

4. THE CORRECTION PROCESS

The user corrects the erroneous actions in the prediction result. This process consists of the judgment of the prediction result, and the conversion for the input of action. The user judges whether the prediction result is correct or not. When the prediction result has errors, the user chooses the erroneous action. According to the corrected action of a user. Values of the changed state of the room are Table 3. About states other than this, the state of the element in the rule changes as it is.

5. EVALUATION EXPERIMENT

The system based on this method has been developed for an experiment to evaluating the adaptability of this method for a user.

[*3] In the analysis of neighboring action probability, the probability that B will happen under A

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{n(A \cap B)/n(\Omega)}{n(A)/n(\Omega)} = \frac{n(A \cap B)}{n(A)}$$

Table 3 Given Values for the Corrected Action

ELEMENT	ACTION	VALUES
Brightness	Don (Doff)	±50 lux
Brightness	Con (Coff)	±250 lux
Brightness	Lon (Loff)	±500 lux
Air conditioner	X°C	X°C
Temperature	Ton (Toff)	±1

5.1 DATA AND PROCEDURE

The data used in this experiment is based on the actual life of the first author. The number of inputted states is 945. The experiment period was nine weeks. In learning process, we consider a range of the natural elements, for example 10 minutes of time, 50Lux of brightness and 2 degrees of temperature, as the judgment standard of the different part and the common part. In the formula (1) the parameters α, β, γ were obtained in the preliminary experiment that is 100, 1 and 10 respectively. The rule dictionary and the state's history are empty at first. The procedure of this experiment is shown in Figure 1.

5.2RESULTS

Figure 2 shows the results in the rate of the correct prediction. In Figure 2, the rate of the correct prediction is increasing as the number of the inputted room's states is increasing. The correct rate is about 83[%] in the final. The rate of week average prediction was 60.1[%]. Table 4 shows the rate of a weekly average positive prediction.

Table 4 Rate of Weekly Average Prediction

Weeks	1	2	3	4	5	6	7	8	9
CR (%)	26.3	33.5	63.0	64.2	68.3	71.8	53.3	77.5	82.8

5.3 CONSIDERATIONS

In Table 4, we can find that the rate of the correct prediction for the 7th week is 53[%]. It is lower than the 6th week (71.5[%]) and the 8th week (77.5[%]). In order to analyze this result, the degree of similarity for data was expressed as Sim(Tx,Ty) and defined as:

$$Sim\ (T_x, T_y) = \frac{T_x \bullet T_Y}{|T_x||T_y|} = \frac{T_x \bullet T_y}{\sqrt{T_X \times T_y}} \bullet\bullet\bullet (5)$$

,where the Sim(Tx,Ty) is the degree of similarity, Tx and Ty are the number of the data for one week used as each evaluation. $T_x \bullet T_y$ is the number of the data that is carrying out that less than in accordance with the given range to the state of natural element in data and the state of the artificial element in data are the same. The analysis result of the degree of similarity used the data of the 7th week to calculate it to the others were shown as the Table 5:

Table 5: the Degree of Similarity (between the 7th week and the others)

Weeks	1	2	3	4	5	6	8	9
Sim (%)	35.6	46.7	50	53.3	42.8	46.7	76.3	78.5

From Table 4 and 5, it was shown that the system in this method carries out dynamic adaptation to a user. That is to say, the system can be learned according to custom and taste of the user. Although the rate of right prediction fell once according to a user's new life rule are learned. It is clear that the rate of correct prediction increased gradually.

6. CONCLUSIONS

In this paper, we proposed the new method---action prediction of a user using Inductive Learning with N-gram. This method acquires rules from the state history automatically by its learning capability. It is not necessary to prepare the rule united or a lot of examples for the user beforehand. It is learned from the example that the system was gained. The rule is acquired according to custom and taste of a user by its own adaptability. Predicting a user's action to some extent was shown. Therefore, the system based on this method can adapt to user automatically, and the result of the experiment shows that this method is effective for the user. The rate of average correct prediction was 60.1[%].

Future works consist of the following contents: the check of the adaptation to a different user, the check of the flexibility of the system using two or more users, automatic adaptation of the range in the natural element's state and use of the Maximum Entropy Model.

7. ACKNOWLEDGEMENT

This work is partially supported by the Grants from the Government subsidy for aiding scientific researches (No.14658097) of the Ministry of Education, Culture, Sports, Science and Technology of Japan.

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